

HAMAR II, Norway

WMO: 013860

Lat: 60.801N

Lon: 11.098E

Elev: 433

StdP: 14.47

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.6	(c) -0.4	(d) -8.7	(e) 3.5	(f) -4.6	(g) -4.3	(h) 4.5	(i) -0.2	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	80.3	61.3	76.3	60.1	72.8	58.6	64.1	74.5	62.6	71.5	61.0	69.7	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.5	80.0	66.8	58.8	75.2	65.9	57.3	71.1	64.6	29.5	74.6	28.4	71.9	27.3	69.7	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) N/A	(b) N/A	(c) N/A	(e) -8.2	(f) 83.8	(g) 7.6	(h) 3.1	(i) -13.6	(j) 86.0	(k) -18.0	(l) 87.8	(m) -22.3	(n) 89.5	(o) -27.8	(p) 91.8		
(4) N/A	(5) N/A	(6) N/A	(e) -9.8	(f) 66.1	(g) 7.8	(h) 2.2	(i) -15.4	(j) 67.7	(k) -19.9	(l) 69.0	(m) -24.3	(n) 70.3	(o) -29.9	(p) 71.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.6	20.3	23.7	31.8	40.7	51.6	58.4	63.1	59.4	52.4	41.1	32.2	23.2		
	DBStd	16.48	9.72	9.33	8.45	6.32	7.86	4.74	5.30	4.18	5.11	6.35	7.89	11.09		
	HDD50	4275	922	737	564	287	76	2	0	1	34	286	535	831		
	HDD65	8613	1387	1157	1029	730	422	204	103	179	379	742	985	1296		
	CDD50	1199	0	0	0	8	124	255	407	291	105	9	0	0		
	CDD65	63	0	0	0	0	5	8	45	5	0	0	0	0		
	CDH74	579	0	0	0	0	73	76	403	27	0	0	0	0		
	CDH80	106	0	0	0	0	7	8	91	0	0	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	20.8	1.1	0.9	1.0	1.0	1.5	2.3	2.9	2.7	2.4	2.0	1.7	1.2		
(15) Precipitation	PrecMax	28.5	2.8	2.8	2.8	2.6	3.0	5.6	6.5	5.7	4.6	3.8	3.0	2.8		
	PrecMin	14.8	0.0	0.1	0.0	0.0	0.3	0.2	0.9	0.4	0.6	0.3	0.5	0.2		
	PrecStd	3.6	0.7	0.6	0.6	0.7	0.8	1.4	1.2	1.5	1.1	1.1	0.8	0.7		
	PrecStd	3.6	0.7	0.6	0.6	0.7	0.8	1.4	1.2	1.5	1.1	1.1	0.8	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.9	43.8	55.6	67.0	80.8	80.8	86.0	77.0	70.6	62.1	52.4	46.4		
		MCWB	36.5	38.1	44.1	50.7	59.1	59.8	64.2	61.9	56.9	55.5	47.8	41.9		
	2%	DB	38.3	40.6	50.7	61.9	75.4	75.9	82.8	74.1	67.6	57.2	48.2	43.0		
		MCWB	35.7	37.3	41.3	48.2	56.7	58.3	63.1	59.7	56.6	50.6	44.6	39.2		
	5%	DB	35.9	37.7	47.0	57.1	70.5	71.9	79.0	71.4	64.8	54.3	45.5	40.1		
		MCWB	33.6	35.2	39.0	46.0	55.3	56.4	61.7	58.7	55.5	48.9	42.7	36.6		
	10%	DB	33.8	36.1	43.4	52.6	65.6	68.7	74.6	68.7	61.9	51.4	42.4	37.2		
		MCWB	31.9	33.9	37.2	42.5	53.3	54.7	61.0	57.7	54.0	46.6	40.2	34.6		
	0.4%	WB	38.3	39.9	45.0	52.0	61.8	63.3	67.0	64.7	60.5	56.2	48.7	42.7		
		MCDB	39.6	42.7	53.6	65.1	73.3	70.8	78.8	71.2	65.4	61.0	51.8	45.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.0	37.1	42.1	48.9	59.3	61.1	65.1	62.8	58.6	52.2	45.5	39.5		
		MCDB	37.8	39.7	49.5	60.2	71.4	70.4	77.3	69.8	64.3	56.0	48.0	42.3		
	5%	WB	34.0	35.5	39.9	46.2	56.7	59.3	63.6	61.3	56.8	49.7	43.0	36.8		
		MCDB	35.6	37.5	45.5	55.4	66.5	68.6	75.0	68.0	62.9	53.3	45.1	39.2		
	10%	WB	32.1	34.0	37.7	43.6	54.6	57.4	62.1	59.6	55.0	47.3	40.3	34.8		
		MCDB	33.7	35.9	43.1	50.9	63.5	66.2	71.9	65.8	60.7	50.8	42.2	36.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	13.1	16.6	18.4	17.8	17.1	17.6	16.6	16.2	12.9	8.9	10.7		
		MCDBR	11.2	13.7	20.0	26.1	25.4	22.9	24.1	21.2	20.8	15.3	13.2	13.4		
		MCWBR	9.5	10.4	12.4	13.9	10.9	10.2	9.1	9.8	12.0	10.6	10.3	11.0		
	5% WB	MCDBR	11.0	12.1	19.1	24.2	21.2	18.7	20.0	15.8	17.4	12.9	12.0	11.9		
		MCWBR	9.8	9.4	12.5	13.4	9.6	9.8	8.5	8.4	10.9	10.1	9.9	10.1		
	Clear-Sky Solar Irradiance	taub	0.235	0.281	0.311	0.351	0.355	0.347	0.366	0.358	0.325	0.310	0.254	0.203		
(40) All-Sky Solar Radiation	RadAvg	taud	2.176	2.274	2.296	2.332	2.428	2.474	2.437	2.454	2.529	2.485	2.263	2.067		
		Ebn at Noon	168	227	258	267	276	280	271	264	255	218	160	126		
	RadStd	Edh at Noon	12	20	28	33	32	31	32	29	23	17	11	8		
		RadAvg	87	292	766	1243	1507	1704	1535	1198	785	348	112	53		
		RadStd	15	56	77	146	152	147	200	107	94	66	30	4		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (56 neighbors)	N/A	N/A	+2.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page